

Design Pattern Explanation - Prepare a one-page summary explaining the MVC (Model-View-Controller) design pattern and its two variants. Use diagrams to illustrate their structures and briefly discuss when each variant might be more appropriate to use than the others.

The MVC design pattern is a software architecture pattern that separates an application into three main components: Model, View, and Controller, making it easier to manage and maintain the codebase. It also allows for the reusability of components and promotes a more modular approach to software development.

Core Components:

- **Model:** Manages the data and business logic.
- **View:** Handles the display and user interface.
- **Controller:** Responds to user input, updates the model and view.

Variants of MVC

1. Model-View-Presenter (MVP)

- **Presenter** replaces the controller and has more control over the view.
- The **View is passive**, forwarding events to the presenter.

[User]

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[View] ↔ [Presenter] ↔ [Model]

- The **Presenter** updates the View explicitly.
- Best for **Windows Forms or Android development**, where unit testing the UI logic is essential.

2. Model-View-View Model (MVVM)

- The **View Model** binds the model to the view using **data binding**.
- The **View Model** exposes data and commands.

[View] ↔ [View Model] ↔ [Model]

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Data Binding

- Ideal for **WPF, Xamarin, Swift UI**, or any UI framework that supports two-way binding.